

Estimated final power budget for ESP32-based propulsion module. Values are estimates.

Assumption	Value	Unit	Notes
Input supply voltage	12.00	V	Original design input rail
Logic rail voltage	3.30	V	For ESP32 and logic-side motor-driver power
Motor rail voltage	6.00	V	For propulsion motors
3.3 V regulator capacity	2.00	A	AP63203WU-7 maximum output current class
6 V regulator capacity	10.00	A	LMZ23610TZ/NOPB maximum output current class

Rail	Component / Load	Qty	Voltage (V)
3.3 V	ESP32-S3-WROOM-1-N4	1	3.300
3.3 V	TB67H450FNG logic side	1	3.300
3.3 V	Status LEDs	2	3.300
3.3 V	UART/debug pullups and small logic	1	3.300
6 V	DC propulsion motor	2	6.000
6 V	Motor startup/stall allowance	2	6.000
6 V	TB67H450FNG motor side	1	6.000

Rail Summary	Estimated Current (A)	Available Current (A)	Margin (A)
3.3 V logic rail	0.265	2.000	1.735
6 V motor rail - normal running	0.820	10.000	9.180

6 V motor rail -
startup/stall case

3.020 10.000 6.980

Final Conclusions

The 6 V motor rail had enough estimated current margin and worked on the final board. The 3.3 V rail was architecturally necessary, but the final PCB implementation failed due to a regulator layout/routing issue. The ESP32 was therefore tested with a breadboard/devkit while the PCB was used where possible for motor-power and driver testing. Encoder feedback was not included in the final power budget as an active load because it was not used in the final implementation.

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ates for report documentation, not final measured current data.

Current Each (A)	Total Current (A)	Power (W)	Status
0.240	0.240	0.792	Designed
0.005	0.005	0.017	Designed
0.005	0.010	0.033	Designed
0.010	0.010	0.033	Designed
0.400	0.800	4.800	Worked
1.500	3.000	18.000	Worst Case
0.020	0.020	0.120	Worked

**Estimated Power
(W)**

Final Test Result

Did not work on
final PCB due to
regulator/layout

0.875 issue

Worked and
4.920 powered motors

Used as
conservative
design margin
18.120 estimate



Notes

Controller logic load estimate

Logic-side input current estimate

Two small indicator LEDs

Allowance for small logic/interface loads

Estimated normal running current per motor

Conservative transient current estimate

Driver overhead estimate, motor current counted separately